| | | | |
|---|---|---|---|
| **Fiscal Year:** | FY 2015 | **Task Last Updated:** | FY 11/19/2014 |
| **PI Name:** | Long, Marshall Ph.D. | | |
| **Project Title:** | Coflow Laminar Diffusion Flame | | |
| **Division Name:** | Physical Sciences | | |
| **Program/Discipline:** | COMBUSTION SCIENCE | | |
| **Program/Discipline--Element/Subdiscipline:** | COMBUSTION SCIENCE--Gaseous diffusion flames | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
| **PI Email:** | marshall.long@yale.edu | **Fax:** | FY |
| **PI Organization Type:** | UNIVERSITY | **Phone:** | 203-432-4229 |
| **Organization Name:** | Yale University | | |
| **PI Address 1:** | Department of Mechanical Engineering | | |
| **PI Address 2:** | PO Box 208284 | | |
| **PI Web Page:** | | | |
| **City:** | New Haven | **State:** | CT |
| **Zip Code:** | 06520-8284 | **Congressional District:** | 3 |
| **Comments:** | | | |
| **Project Type:** | Flight | **Solicitation / Funding Source:** | 2001 Physical Sciences 01-OBPR-08 |
| **Start Date:** | 01/01/2012 | **End Date:** | 12/31/2016 |
| **No. of Post Docs:** | | **No. of PhD Degrees:** | 1 |
| **No. of PhD Candidates:** | 1 | **No. of Master' Degrees:** | |
| **No. of Master's Candidates:** | | **No. of Bachelor's Degrees:** | |
| **No. of Bachelor's Candidates:** | | **Monitoring Center:** | NASA GRC |
| **Contact Monitor:** | Stocker, Dennis P | **Contact Phone:** | 216-433-2166 |
| **Contact Email:** | dennis.p.stocker@nasa.gov | | |
| **Flight Program:** | ISS | | |
| **Flight Assignment:** | ISS<br>NOTE: Extended to 12/31/2016 per D. Stocker/GRC and NSSC information (Ed., 1/14/2015)<br>NOTE: change in Period of Performance to 1/1/2012-12/31/2014 per NSSC information (previous POP was 10/1/11-9/30/14)--Ed., 12/30/2011 | | |
| **Key Personnel Changes/Previous PI:** | None | | |
| **COI Name (Institution):** | Smooke, Mitchell Ph.D. ( Yale University ) | | |
| **Grant/Contract No.:** | NNX11AP43A | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |

| | |
|---|---|
| **Task Description:** | Editor's Note: Successor agreement to Fuel Dilution Studies of Coflow Laminar Diffusion Flames in a Microgravity Environment (NNX07AB33A), per D. Stocker, NASA Glenn Research Center.<br>Diffusion flames are the flame type of most practical combustion devices. The ability to predict the coupled effects of complex transport phenomena with detailed chemical kinetics in such systems is critical in the modeling of turbulent reacting flows and in understanding the processes by which extinction, soot formation, and radiative transfer take place. Turbulent diffusion flames are generally nonbuoyant due to the large flow velocities. In addition, in practical devices combustor configuration and flow field patterns may produce flames with sharp edges. Extinction along these edges can result from incomplete combustion. In other regions combustion generated soot particulates may be formed. To understand better the factors that affect flame extinction and particulate formation, we propose to focus this study on flame phenomena at the extremes of the fuel dilution spectrum – namely dilute flames near extinction and pure-fuel (sooting) flames. Previous work has shown that existing computational models are unable to predict accurately the behavior of these systems. The enhanced time and spatial scales in a microgravity environment will enable the stabilization of flames with increased levels of dilution and will help generate flames that have increased soot residence times compared to normal gravity facilities. This should provide an excellent platform to gain a better understanding of the factors that affect diffusion flame lift-off distance and extinction, and it should also provide an excellent test-bed to determine the roles of inception, surface growth, and oxidation processes in soot formation models. The overall goals of the research are to improve our understanding of the physical and chemical processes controlling diffusion (i.e., non-premixed) flame structure and lifting phenomena (i.e., stabilization) and to provide for rigorous testing of numerical models, including thermal radiation, soot formation, and detailed chemical kinetics. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | Combustion of fossil fuels is responsible for the majority of energy production in the world today and provides the power for transportation systems of all kinds. Effective modeling of combustion processes is critical to improving efficiencies and reducing pollutants in practical combustion devices. The ability to predict the coupled effects of complex transport phenomena with detailed chemical kinetics in diffusion flames will improve modeling of turbulent reacting flows and in understanding the processes by which extinction, soot formation, and radiative transfer take place. |
| **Task Progress:** | Quantitative Measurements with Simple Diagnostics<br>We have continued the refinement and characterization of diagnostics and flame configurations that will allow us to take greatest advantage of the long periods of high-quality microgravity that are available on the International Space Station (ISS). We have worked to develop techniques for making the comparison between experiments and computations as quantitative as possible, even with relatively simple diagnostic techniques. A technique utilizing a color digital camera has been developed that allows observation of flame shape, size, lift-off height, and measurements of temperature via soot and thin-filament pyrometry as well as quantitative determination of the soot volume fraction. Techniques for obtaining an absolute light calibration have also been developed, which is essential for measuring the absolute soot volume fraction and the number density of chemiluminescent species such as CH*. In an effort to improve the diagnostics, thin-filament pyrometry has recently been assessed against laser-based techniques. Several sources of error were identified and uncertainty analysis has been performed to estimate the measurement uncertainty and provide insight on error reduction.<br><br>The color camera to be used in the ACME experiments has been specified and we have purchased a copy of the detector. The spectral response and linearity of the detector have been characterized, which is essential for accurate temperature, CH*, and soot volume fraction measurements.<br><br>Microgravity Experiments: SLICE and CLD Flame<br><br>As a precursor to CLD Flame, the Structure & Liftoff in Combustion Experiment (SLICE) took advantage of the existence of a closely related set of experiments recently completed on board the ISS. The goal of SLICE was to investigate the structure of lifted and sooty flames. The SLICE experiments were carried out on board the ISS early in 2012. A comprehensive data set with a variety of test conditions, including different hydrocarbon fuels with varying dilution levels, different fuel nozzle sizes and various flow conditions, was obtained. A fully characterized color digital camera was used to image line-of-sight flame emissions to obtain flame properties such as the flame shape and size, lift-off height, as well as soot temperature and volume fraction. The data analysis that has been done includes quantitative determination of soot temperature and volume fraction for all sooty flames and lift-off height measurements for lifted flames. Similar tests using a copy of the SLICE hardware under normal gravity have also been performed, which enables direct comparisons of these flame properties between 0-g flames and 1-g flames. Numerical simulation of some of the flames has also been performed and compared with experimental results. This work resulted in two presentations at the 35th International Symposium on Combustion and two papers in the Proceedings of the Combustion Institute. Furthermore, the results provided important data for continued development of the computational model and soot sub-model, and gave experimental insights that will help minimize risk and maximize the scientific return of the CLD Flame Experiment.<br><br>The CLD Flame hardware is currently being developed, and we have worked to improve the design of the coflow burner. A series of refinements were made to the fuel nozzle and the flow straightener used in the coflow. Detailed velocity measurements were made, which will provide the boundary conditions needed for the computational modeling. The improved coflow provided lifted flames with a higher degree of symmetry than previous designs. Following construction and characterization of the new burner at Yale, it was shipped to NASA Glenn for testing in their drop tower facilities.<br><br>High Dynamic Range (HDR) Imaging<br><br>SLICE yielded thousands of images of coflow diffusion flames in a microgravity environment under different conditions. The database includes an assortment of microgravity flames that vary in liftoff height, soot load, and shape, depending on fuel type (CH4 or C2H4), dilution level (with N2), nozzle diameter, and gas velocity. SLICE employed a Nikon D300s digital camera as the principal diagnostic tool. Microgravity flames have been found to be sootier than their 1 g counterparts. Over a series of exposures of sooty flames, this often resulted in some saturated images that, by themselves, are not useful for quantitative measurements. Until recently, this limited the analysis to a subset of the SLICE data. However, development and testing of High Dynamic Range (HDR) imaging algorithms has proved successful in overcoming these limitations. By combining information from multiple exposures of the same flame (even |

partially saturated images), the overall signal/noise of the measurement can be increased, resulting in improved temperature and soot volume fraction measurements. Given the high number of unstudied cases left in the database, the new processing techniques will provide useful data for additional investigations. Moreover, these new approaches will likely be beneficial for flame analysis in ACME.

Recent Computational Progress

The computational models used in our earlier flame studies employed a velocity-vorticity fluid dynamic approach where the Navier-Stokes equations were replaced by two nonlinear Poisson equations and a vorticity transport equation for the axial and radial velocities and the vorticity, respectively. While excellent agreement with experimental results has been obtained for numerous coflow flame studies, there are cases in which the agreement is less satisfactory. These are typically flames in which large vorticity values are generated near the base of the flame. The result is that the flames can have small regions of mass loss or gain. This results from the fact that in a velocity-vorticity model, it is the gradient of the continuity equation that is solved to machine zero and not the continuity equation itself. To improve mass conservation, we have recently investigated modifications to our previous velocity-vorticity formulation whereby vorticity terms in the velocity equations are replaced with modified difference expressions. Mass loss/gain is reduced dramatically in these problems.

We have also continued our studies of sooting flames in both a normal gravity and a microgravity environment. Despite the complexities of modeling soot formation in flames using detailed chemistry, the linkage of soot production to radiation and bulk flame properties is so strong that the coupled treatment of this problem has become a necessity for quantitative modeling of flame structure. Refinements to an earlier version of our soot model have recently been implemented. In this work the soot volume fraction is predicted using a sectional representation where the model assumes that soot particles are approximated as carbon spheres and that they exist in the free molecular limit. Rather than letting the surface growth process continue without any restriction on the size of the particles, we limit the size of the primary particles to approximately 25 nm. A further assumption is that within a given section, spheroids of varying diameter do not exhibit vastly differing aerosol dynamical qualities. This assumption holds as long as the number of sections is sufficiently large. In our most recent computations, a total of 30 soot sections were employed.

| **Bibliography Type:** | Description: (Last Updated: 01/12/2026) |
|---|---|
| **Articles in Peer-reviewed Journals** | Ma B, Wang G, Magnotti G, Barlow RS, Long MB. "Intensity-ratio and color-ratio thin-filament pyrometry: Uncertainties and accuracy." Combustion and Flame. 2014 Apr;161(4):908-16. http://dx.doi.org/10.1016/j.combustflame.2013.10.014 (Previously reported as "Available online 15 November 2013. In press, corrected proof.") , Apr-2014 |
| **Articles in Peer-reviewed Journals** | Ma B, Long MB. "Combined soot optical characterization using 2-D multi-angle light scattering and spectrally resolved line-of-sight attenuation and its implication on soot color-ratio pyrometry." Applied Physics B. 2014 Oct;117(1):287-303. http://dx.doi.org/10.1007/s00340-014-5834-x , Oct-2014 |
| **Articles in Peer-reviewed Journals** | Ma B, Cao S, Giassi D, Stocker DP, Takahashi F, Bennett BAV, Smooke MD, Long MB. "An experimental and computational study on soot formation in a coflow jet flame under microgravity and normal gravity" Proceedings of the Combustion Institute. Available online 25 June 2014. In press, Corrected Proof. http://dx.doi.org/10.1016/j.proci.2014.05.064 , Jun-2014 |
| **Articles in Peer-reviewed Journals** | Cao S, Ma B, Bennett BAV, Giassi D, Stocker DP, Takahashi F, Long MB, Smooke MD. "A computational and experimental study of coflow laminar methane/air diffusion flames: Effects of fuel dilution, inlet velocity, and gravity." Proceedings of the Combustion Institute. Available online 26 June 2014. In press, Corrected Proof. http://dx.doi.org/10.1016/j.proci.2014.05.138 , Jun-2014 |